

Daily Number Theory Drill #3

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: *Cyclic Patterns · Telescoping Exponents · Last Digit Sequences.*

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition

(≤15 s each)

Remainders, parity, basic prime factors, and pure recognition.

- Find the remainder when $10^{10} + 10^5 + 1$ is divided by 9.
- What is the unit digit of 3^{2026} ?
- Find the largest prime factor of 1001.
- Find the remainder when $7 \times 8 \times 9$ is divided by 5.
- If n is an integer such that $2n \equiv 3 \pmod{5}$, what is the smallest positive value of n ?
- Evaluate $\gcd(144, 84)$.
- Find the number of positive divisors of 36.
- Find the remainder when 2^{10} is divided by 11.
- Express 120 as a product of prime factors.
- Find the smallest positive integer k such that $12k$ is a perfect square.

Section B Manipulation Drills

(≤50 s each)

Divisibility rules, modulo properties, algebraic factoring over integers.

- How many of the numbers 3, 4, 5, 6, 7 are factors of the sum $3^{2026} + 3^{2025} + 3^{2024}$? (after SMC 2023 Q10)
- Find the highest power of 2 that divides $100!$.
- If a and b are non-zero digits, find the maximum possible integer value of $\frac{10a+b}{a+b}$.
- Find the number of integer pairs (x, y) such that $xy = 24$.

5. What is the smallest positive integer greater than 2 that leaves a remainder of 2 when divided by 3, 4, and 5?
6. Find the remainder when $11^{11} + 12^{12}$ is divided by 10.
7. Find the value of p if p and $p^2 + 2$ are both prime numbers.
8. Given that n is an integer, for how many values of n is $\frac{2n+15}{n+2}$ an integer?
9. Find the sum of all positive integers n for which $n^2 - 1$ is a prime number.
10. Determine the two last digits of $5^{2026} + 6^{2026}$.

Section C Substitution & Structure

(≤90 s each)

Link Number Theory to Algebra and Combinatorics. Look for patterns.

1. Find all pairs of positive integers (x, y) such that $\frac{1}{x} + \frac{1}{y} = \frac{1}{3}$.
2. Solve the equation $x^2 - y^2 = 2025$ for positive integers (x, y) such that y is as small as possible.
3. Find all pairs of integers (x, y) that satisfy $xy + x + y = 14$.
4. Find all integers x such that $x^4 + x^2 + 1$ is a prime number.
5. How many ways can we choose the digits a and b such that the 5-digit number $3a4b2$ is divisible by 36?
6. Find the number of 3-element subsets of $\{1, 2, \dots, 10\}$ such that the sum of the elements is divisible by 3. *(after MAT combinatorics)*
7. Find the sum of all integer values of x for which $\frac{x^2 + 7x + 2}{x + 1}$ is an integer.
8. Find the last two digits of 2019^{2025} . *(after SMC binomial last digits)*

Section D Challenge

(TMUA / SMC / BMO1 difficulty)

Insight over computation. Each question has a short, elegant solution — find it.

1. Find the number of pairs of positive integers (a, b) with $a \leq b$ such that $\text{lcm}(a, b) = 2^3 \cdot 3^2 \cdot 5$. *(after BMO1 combinatorics)*
2. Find all pairs of prime numbers (p, q) such that $p^2 - 2q^2 = 1$. *(after BMO1 Diophantine)*
3. Find the number of integer solutions to $x^3 + 2y^3 = 4z^3$. *(after BMO1 infinite descent)*
4. Find the last three digits of 7^{9999} . *(after SMC massive exponents)*

5. Find the last three digits of 2025^{2026} . (*after SMC cyclic endings*)

“Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths